

Ensure Data Accuracy and Integrity with LangChain Queries

Data Integrity, Validation, and Error Handling in LangChain

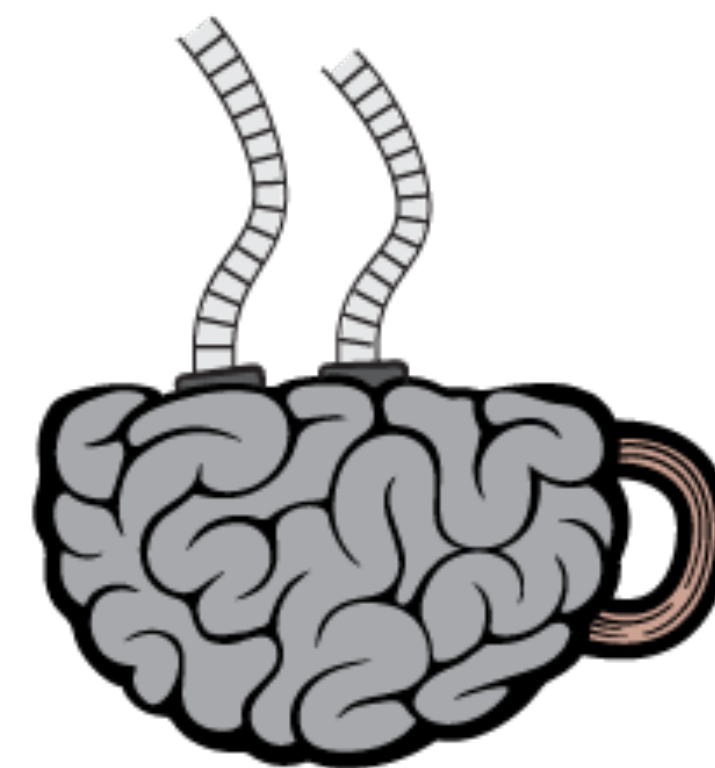


Daryle Serrant

Machine Learning Engineer

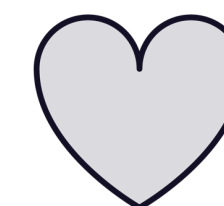
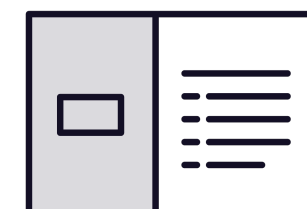
@daryleserrant | www.daryleserrant.com

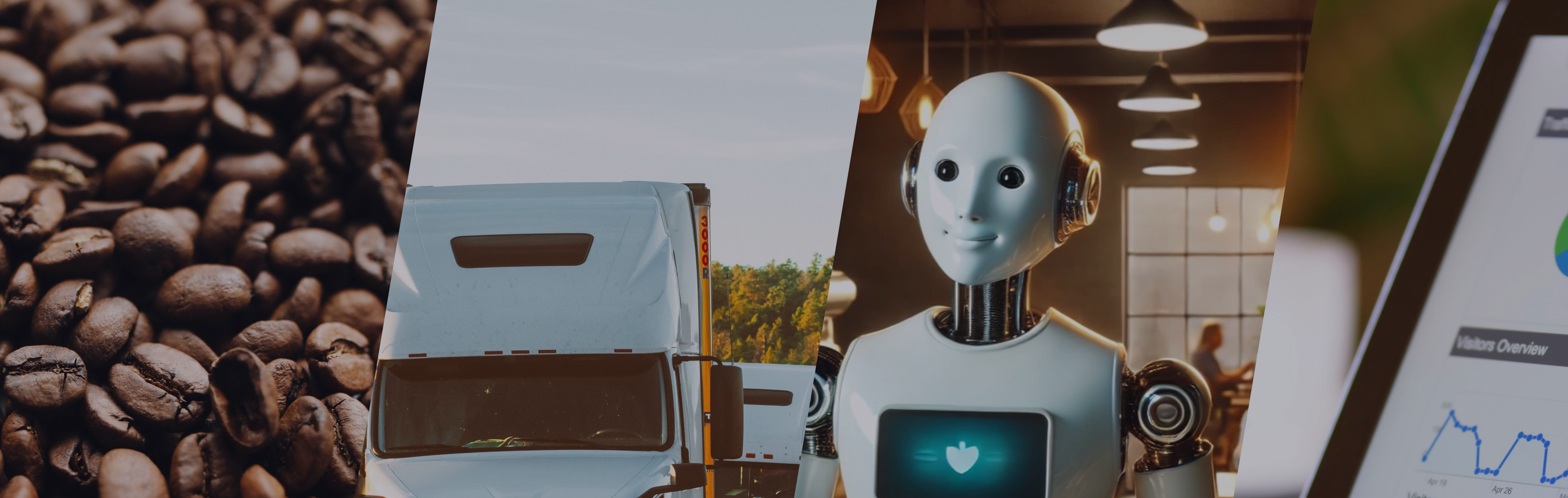




WIRED BRAIN

— COFFEE —



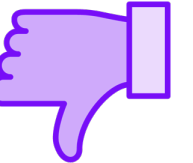


Infusing Generative AI into Coffee Making



The quality of Wired Brain Coffee's data *is not so great*



terrible 

The quality of Wired Brain Coffee's data ~~is not so great~~



Wired Brain Coffee's Data Quality Issues



Inconsistent data entry



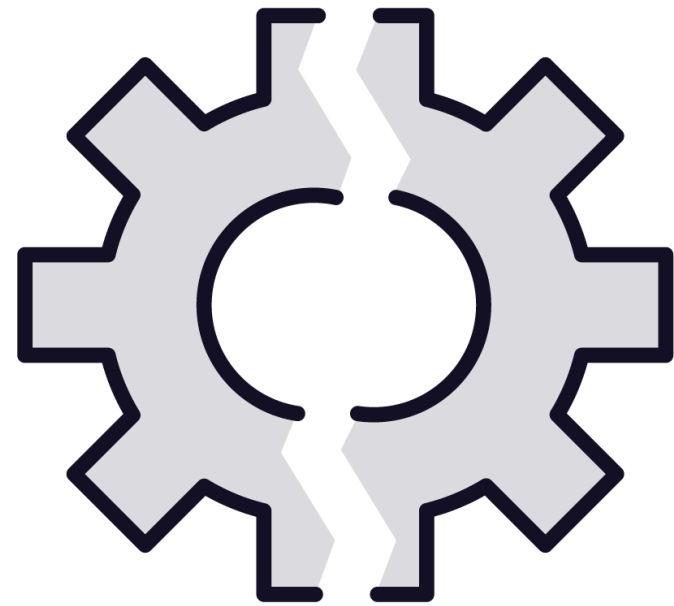
Discrepancies in sales data



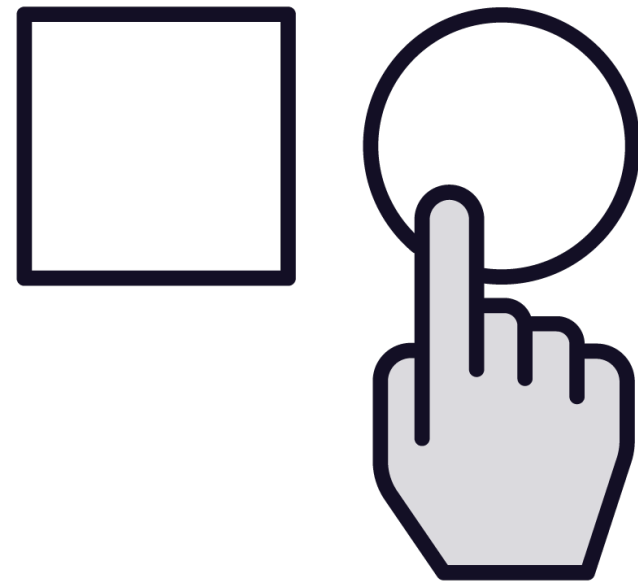
Inaccurate employee performance metrics



Data Quality Risks



**Operational
inefficiencies**



**Poor decision
making**



**Loss of
customer trust**



**Data protection
and privacy
violations**



Data Integrity Features in LangChain

Data validation

Error handling

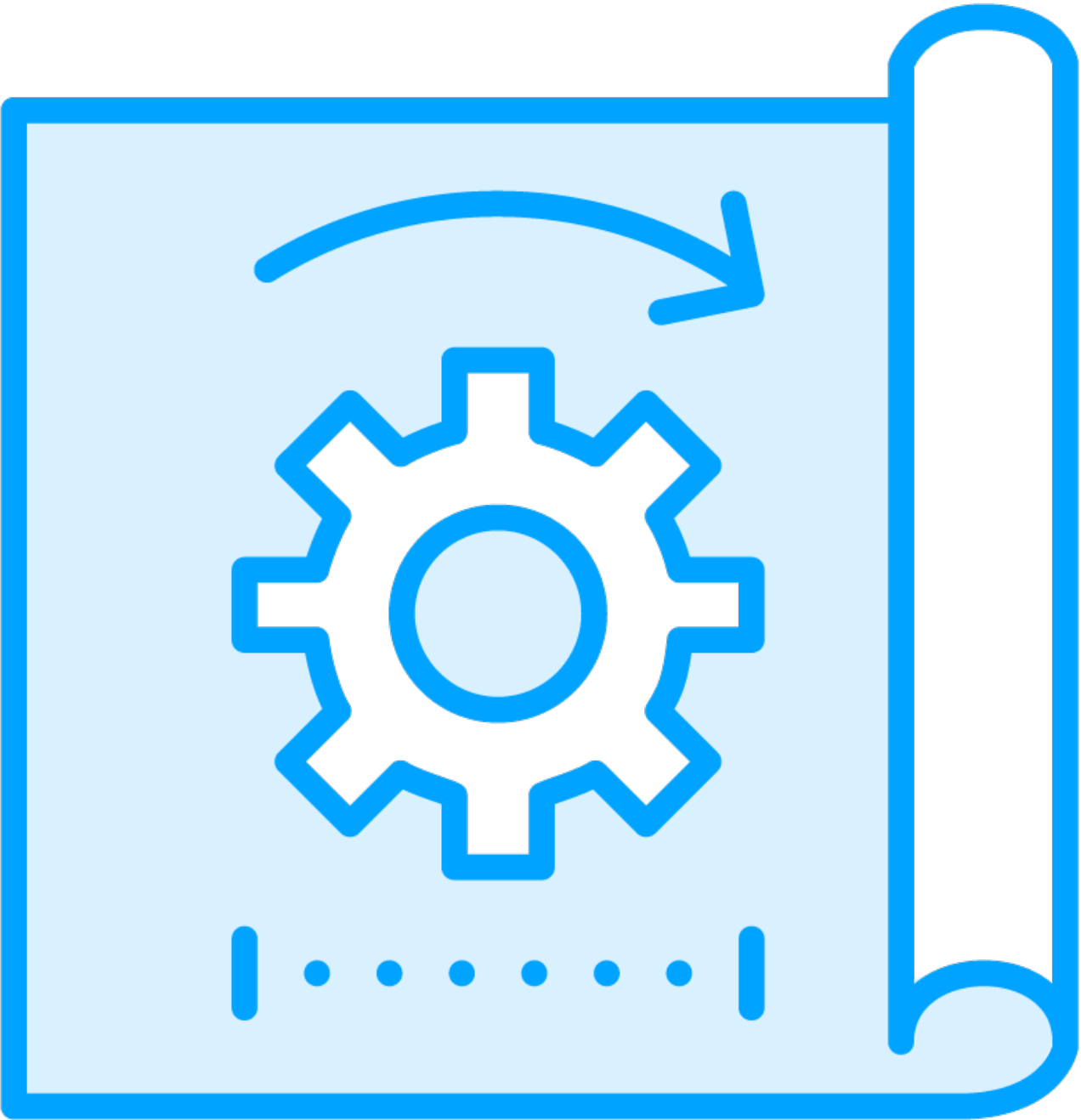
Anomaly detection

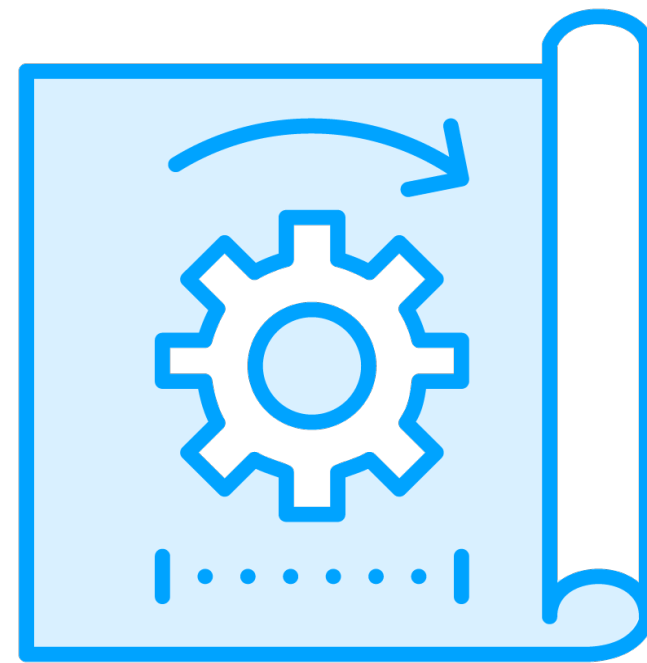
Data privacy and safety



Data Validation and Error Handling

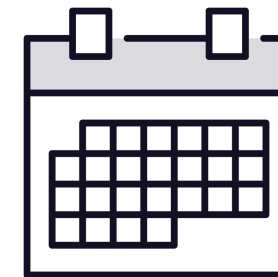
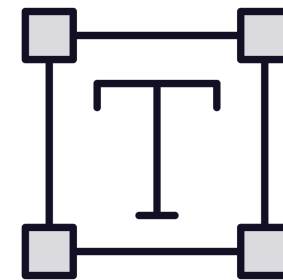
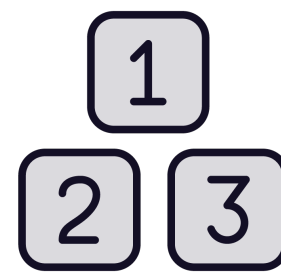






Pydantic

Fast and extensible library for data validation



Validating Sales Transactions Using Pydantic

validate_sale.py

```
from datetime import datetime
from pydantic import BaseModel
from pydantic import ValidationError

class SalesTransaction(BaseModel):
    timestamp: datetime
    product_id: int
    quantity: int
    price: float

try:
    tx = SalesTransaction(timestamp="2024-03-25T16:14:03Z",
product_id="5abc2", quantity=8)
except ValidationError as e:
    print(e.errors())
```

```
> python validate_sale.py
[{'type': 'int_parsing', 'loc': ('product_id',), 'msg': 'Input should be a valid integer, unable to parse string as an integer', 'input': '5abc2', 'url': 'https://errors.pydantic.dev/2.8/v/int_parsing'}, {'type': 'missing', 'loc': ('price',), 'msg': 'Field required', 'input': {'timestamp': '2024-03-25T16:14:03Z', 'product_id': '5abc2', 'quantity': 8}, 'url': 'https://errors.pydantic.dev/2.8/v/missing'}]
```



Fixing Output Parsing Errors

OutputFixingParser

RetryOutputParser



Both modules do not currently work!

Let's see an alternative...





Human-in-the-loop Validation

Ideal for critical or ambiguous data entries



Automated Validation for Customer Birthdays



Inconsistent date formats



Out-of-range dates



Missing data



More Data Quality Strategies



Implement systems for consistent data entry



Use feedback mechanisms to identify issues

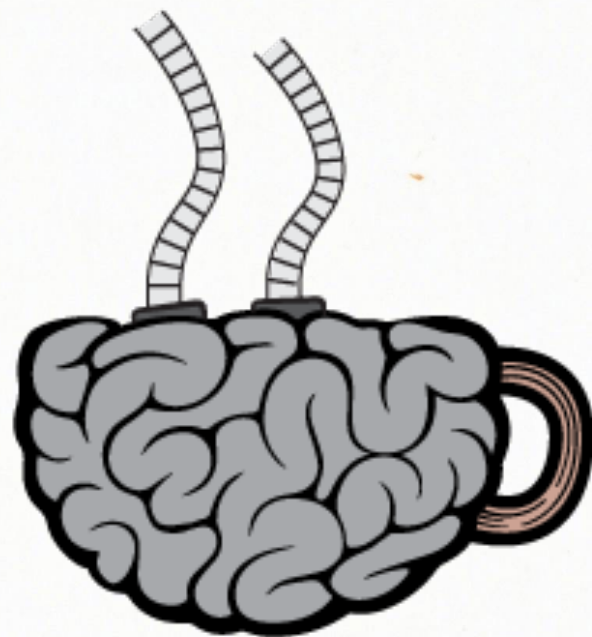


Anomaly Detection Using LangChain





Finding unusual patterns that indicate something is wrong



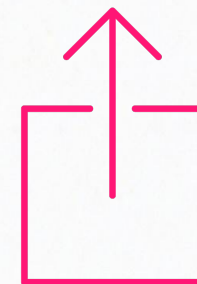
WIRED BRAIN
— COFFEE —



Drop in sales



Pricing issues

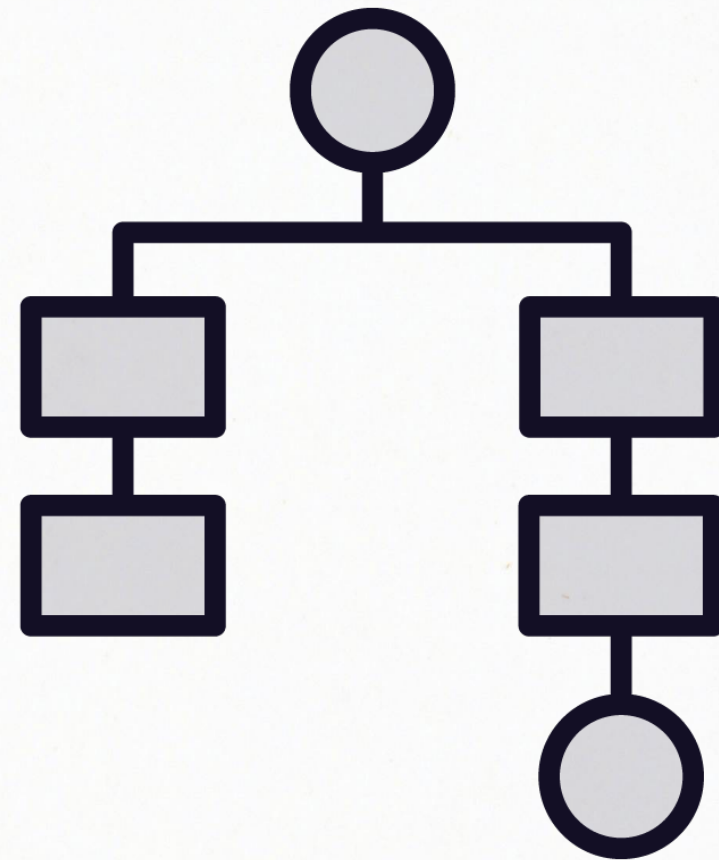


Other external factors



Typical anomaly detection methods depend on specific data formats

Autoencoders
ARIMA
K-nearest neighbors
Local outlier factor
K-means
Support vector machine
Etc.

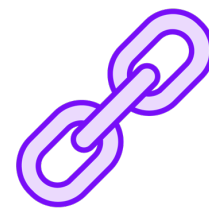


Messy data?
Unstructured formats?

Detecting Anomalies Using LLMs



Prompt engineering



Chain-of-thought prompting

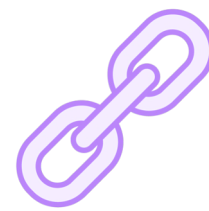
“Describe your reasoning in steps...”



Detecting Anomalies Using LLMs

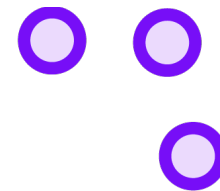


Prompt engineering



Chain-of-thought prompting

“Describe your reasoning in steps...”



Few-shot prompting

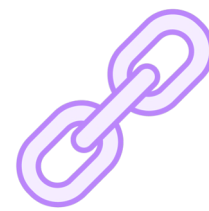
“Here are some examples...”



Detecting Anomalies Using LLMs

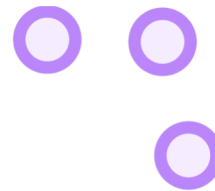


Prompt engineering



Chain-of-thought prompting

“Describe your reasoning in steps...”



Few-shot prompting

“Here are some examples...”



Self-reflection

“Double-check your findings...”



Detecting Anomalies Using LLMs

Identify data points that deviate significantly from peers



Vector databases





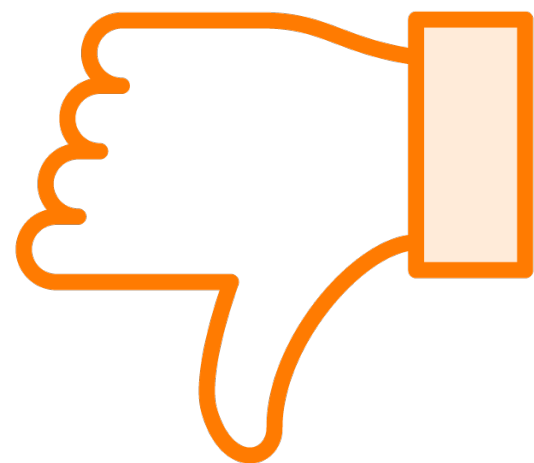
Data Privacy and Safety





Moderation Chain

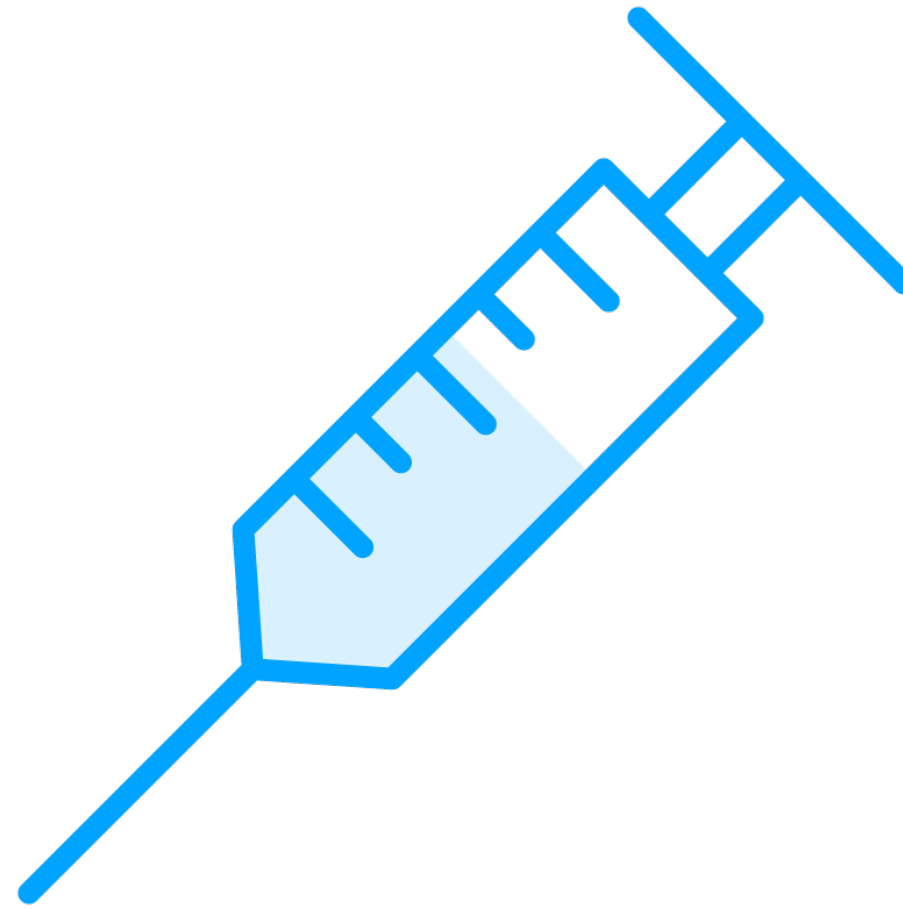
Detects and prevents the generation of harmful content



Constitutional Chain

Ensures that LLM outputs adhere to a set of principles

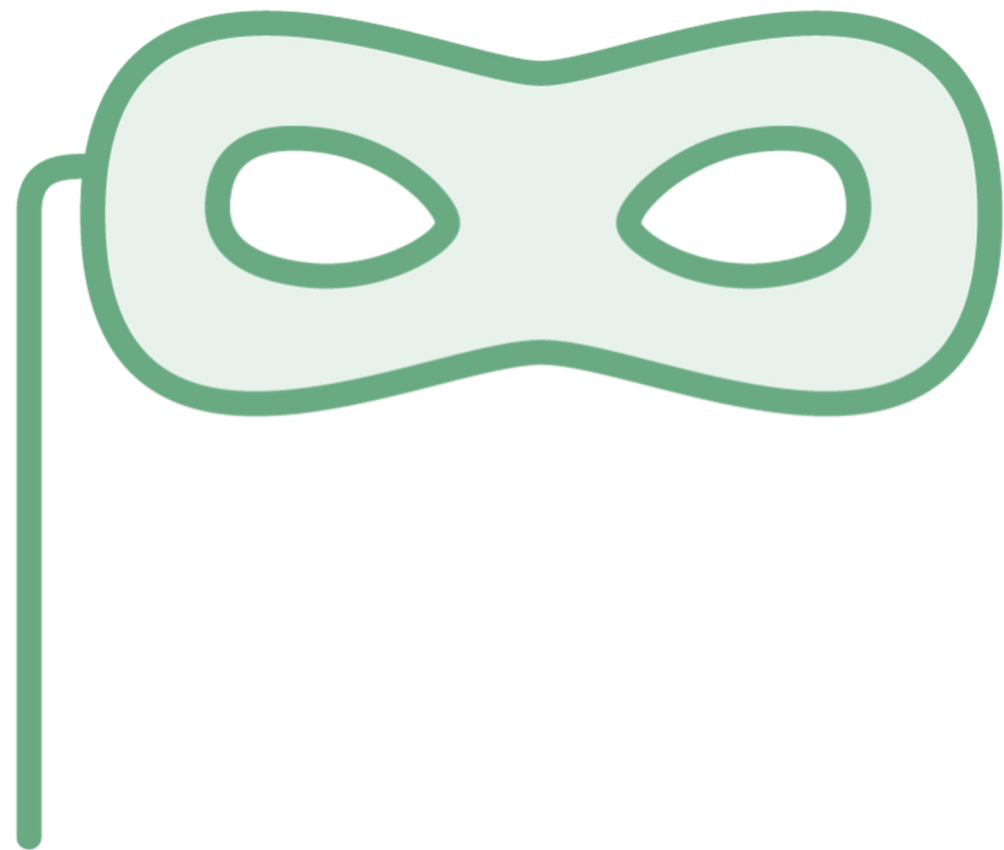




Prompt Injection Identification

Uses text classification models to identify malicious prompts





Data Anonymization

Removes sensitive information from data



How Data Anonymization Works

1ST

Identify fields containing personally identifiable information

2ND

Replace personal information with pseudo values



Data Anonymization Using Microsoft Presidio

More information:

https://python.langchain.com/v0.1/docs/guides/productionization/safety/presidio_data_anonymization/

anonymize_contact.py

```
from langchain_experimental.data_anonymizer import PresidioAnonymizer

anonymizer = PresidioAnonymizer()

anonymizer.anonymize("My name is Marco Polo, call me at 323-775-8321 or email me at marco.polo.jones@gmail.com")
```

```
> python anonymize_contact.py
My name is Donald Ball, call me at (631)549-2726 or email me at benjamin56@example.org
```





Data Encryption

Attackers can access sensitive information stored in vector databases

Encrypt sensitive data in vector databases

